

Day 12 Plan

Maths: Mensuration 2D + Statistics · Reasoning: Statement–Conclusion, Data Sufficiency · Science: Chemistry 4: Carbon Compounds, Fuels & Everyday Chemistry

Today's schedule (about 5½ hours with breaks)

#	Block	Time	What you do	Where
0	Day 11 recall	15 min	10 questions from Day 11, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	5 simplification questions for speed → formulas & methods (10 min) → 2-3 book questions per pattern, timed → 10 Testbook PYQs on this topic → log mistakes	Section A · book · Testbook
–	Break	10 min	Away from the screen	–
2	Reasoning	45 min	Learn the method (10 min) → 15 timed questions, Testbook PYQs first → log	Section B · Testbook
–	Break	10 min	Away from the screen	–
3	Science	110 min	6 lessons with quick checks (≈60 min) → 20-question practice set → Testbook PYQs: carbon compounds, fuels, everyday chemistry	Section C · Testbook
–	Break	10 min	Away from the screen	–
4	General awareness	45 min	20 current-affairs MCQs (20 min) → static GA: Culture — classical & folk dances, festivals, books & authors — Testbook GA PYQs on it (25 min)	CA PDF · Testbook
5	Close	15 min	Redo today's wrong questions · write 3 rules for tomorrow · send me your scores	Mistake log

Short day? Cut in this order: Testbook science PYQs → reasoning down to 10 questions → static GA down to 10 min. Never cut recall, maths or the science lessons. · Exam pace (average 54 sec a question): GA ~30 sec · Science ~45 · Reasoning ~50 · Maths ~65.

Block 0 — Day 11 recall (answers on the last page)

#	Question	Your answer
1	Metal that is liquid at room temperature?	
2	Most ductile metal?	
3	Best conductor of electricity?	
4	Brass is made of?	
5	Thermite reaction is used to?	
6	Main ore of aluminium?	
7	Eka-silicon is now called?	
8	Gas filled in filament bulbs?	
9	Group 17 elements are called?	
10	Maths: ₹40 and ₹60 rice mixed to ₹52 — ratio?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Mensuration 2D + Statistics

Core idea

Mensuration questions are formula + careful units. Draw the figure, mark what is given, and check whether the question wants **area** (square units) or **perimeter** (units). For statistics, **always sort the data first** before finding the median.

Formulas

Concept	Formula / rule
Rectangle	Area lb · perimeter $2(l + b)$ · diagonal $\sqrt{l^2 + b^2}$
Square	Area a^2 · diagonal $a\sqrt{2}$ · area = $d^2/2$
Triangle	$\frac{1}{2} \times \text{base} \times \text{height}$ · Heron: $\sqrt{s(s-a)(s-b)(s-c)}$, $s = (a + b + c)/2$
Equilateral triangle	Area $(\sqrt{3}/4)a^2$ · height $(\sqrt{3}/2)a$
Circle	Area πr^2 · circumference $2\pi r$ · semicircle perimeter $\pi r + 2r$
Sector (angle θ)	Area $(\theta/360)\pi r^2$ · arc $(\theta/360)2\pi r$
Parallelogram · rhombus · trapezium	base $\times$ height · $\frac{1}{2} d_1 d_2$ · $\frac{1}{2}(a + b)h$
Side changes by $x\%$	Area changes by $(2x + x^2/100)\%$
Mean · median · mode	Sum $\div n$ · middle value after sorting ($(n + 1)/2$ th if n odd) · most frequent
Empirical relation	Mode = $3 \text{ median} - 2 \text{ mean}$

Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Rectangle	Pythagoras	$12 \times 5 \rightarrow$ diagonal 13 , area 60
2. Equilateral	$(\sqrt{3}/4)a^2$	Side 6 $\rightarrow 9\sqrt{3} \approx$ 15.59
3. Heron	$s = 21$	13, 14, 15 $\rightarrow \sqrt{(21 \times 8 \times 7 \times 6)} = \sqrt{7056} =$ 84
4. Circle from circumference	$2\pi r \rightarrow r$	44 cm $\rightarrow r = 7 \rightarrow$ area 154 cm²
5. % change	$2x + x^2/100$	Each side +20% $\rightarrow$ area +44%
6. Path outside	Outer – inner	Garden 20×10 m, path 2 m wide outside $\rightarrow 24 \times 14 - 200 =$ 136 m²
7. Rhombus	$\frac{1}{2} d_1 d_2$	Diagonals 16 and 12 $\rightarrow$ area 96 ; side $\sqrt{(8^2 + 6^2)} =$ 10
8. Wheel	Distance $\div$ circumference	Radius 35 cm $\rightarrow 220$ cm per turn $\rightarrow 1.1$ km = 110,000 cm $\rightarrow$ 500 turns
9. Mean, median, mode	Sort first	3, 7, 7, 2, 9, 7, 5 $\rightarrow$ sorted 2, 3, 5, 7, 7, 7, 9 $\rightarrow$ mean $40/7 \approx$ 5.71 , median 7 , mode 7
10. Empirical	$3 \text{ median} - 2 \text{ mean}$	Mean 20, median 22 $\rightarrow$ mode = $66 - 40 =$ 26

Traps

- Radius vs diameter — read twice.
- Semicircle perimeter includes the diameter: $\pi r + 2r$.
- Median: sort first; with an even count, average the two middle values.

B · REASONING — Statement-Conclusion, Data Sufficiency

Statement-conclusion: a conclusion follows only if it comes **directly from the statement**, using no outside knowledge. Be suspicious of extreme words: **only, all, always, never, best**.

Statement-assumption: an assumption is something the speaker **takes for granted** before saying it. An advertisement assumes people read it and can be influenced. A notice assumes people will follow it.

Data sufficiency: you only decide **whether** the question can be answered — don't solve it fully. Usual options: I alone · II alone · either alone · both together · neither.

Patterns, with solved examples

Pattern	Example	Solution
Conclusion	'Regular exercise keeps the body fit.' I: People who exercise regularly are fit. II: Only exercise keeps the body fit.	Only I ('only' is extreme)
Assumption (ad)	'Use XYZ toothpaste for whiter teeth.' I: People want whiter teeth. II: Other toothpastes don't whiten.	Only I
Assumption (notice)	'Passengers are requested to keep their luggage with them.' I: Unattended luggage may be lost or misused. II: Passengers usually travel without luggage.	Only I
DS — both	What is A's age? I: A is twice as old as B. II: B is 10.	Both together
DS — trap	Is x even? I: x is divisible by 4. II: x is a prime number.	I alone (II fails: 2 is prime and even, 3 is prime and odd)
DS — either	Which way is P facing? I: If P turns left, he will face north. II: In the morning, P's shadow falls directly behind him.	Either alone — I: facing east · II: morning shadow points west, so he faces east

Speed rule

Data sufficiency: test each statement alone first, then together. Stop the moment you know whether it **CAN** be answered.

C · SCIENCE — Chemistry 4: Carbon Compounds, Fuels & Everyday Chemistry

6 lessons that finish chemistry. Carbon compounds and fuels were 8 of 210 questions (R), but 'everyday chemistry' (common names, polymers, medicines) shows up across chemistry and GA. Lesson 6 is a high-yield fact table.

Lesson 1 · Why carbon is special; hydrocarbons (about 9 min)

Carbon has valency **4** (tetravalent) and can bond to itself in long chains, branches and rings — **catenation**. That is why there are millions of carbon compounds. Its bonds are **covalent**.

Family	Bond	General formula	First members
Alkanes (saturated)	Single C-C	C_nH_{2n+2}	Methane CH_4 , ethane C_2H_6 , propane C_3H_8 , butane C_4H_{10}
Alkenes (unsaturated)	Double C=C	C_nH_{2n}	Ethene C_2H_4 (ripens fruit), propene
Alkynes (unsaturated)	Triple $C\equiv C$	C_nH_{2n-2}	Ethyne C_2H_2 (acetylene — oxy-acetylene welding)

Calcium carbide (used illegally to ripen fruit) releases **acetylene** with moisture. Ethene is the natural ripening hormone.

Quick check 1 — answer out loud before moving on (answers at the end)

a) General formula of alkenes? b) Gas used with oxygen for welding?

Lesson 2 · Homologous series and functional groups (about 9 min)

A **homologous series** has the same functional group; neighbours differ by **CH_2** (14 u) and have similar chemical properties.

Group	Formula	Example and use
Alcohol	-OH	Ethanol C_2H_5OH — drinks, sanitiser, fuel blending; methanol is poisonous (blindness)
Aldehyde	-CHO	Methanal (formaldehyde); 40% solution = formalin, preserves specimens
Ketone	$>C=O$	Propanone (acetone) — nail-polish remover
Carboxylic acid	-COOH	Ethanoic (acetic) acid; vinegar = 5-8% acetic acid ; pure acid freezes in winter = glacial acetic acid
Halide	-Cl, -Br	Chloroform $CHCl_3$ — old anaesthetic
Ester	-COO-	Sweet-smelling (perfumes, flavours); acid + alcohol → ester + water (esterification)

Quick check 2 — answer out loud before moving on (answers at the end)

a) Which acid is in vinegar, and at what strength? b) Functional group of ethanol?

Lesson 3 · Reactions of carbon compounds; soaps and detergents

(about 10 min)

Reaction	What happens	Example
Combustion	Burns to $\text{CO}_2 + \text{H}_2\text{O} + \text{heat}$	Saturated → clean blue flame · unsaturated → sooty yellow flame
Oxidation	Alcohol → acid	With alkaline KMnO_4 or acidified $\text{K}_2\text{Cr}_2\text{O}_7$
Addition (hydrogenation)	H_2 added across a double bond, nickel catalyst	Vegetable oil → vanaspati ghee
Substitution	H replaced by Cl	Methane + chlorine in sunlight
Saponification	Ester (fat) + NaOH → soap	How soap is made

Soap = sodium or potassium salt of a long-chain fatty acid. Its tail loves oil, its head loves water; molecules form **micelles** that trap grease. In **hard water** (Ca^{2+} , Mg^{2+}) soap forms **scum** and fails; **detergents** work in hard water.

Hardness: **temporary** (bicarbonates) — removed by boiling; **permanent** (chlorides, sulphates) — removed by washing soda.

Quick check 3 — answer out loud before moving on (answers at the end)

a) Catalyst used to turn vegetable oil into vanaspati? b) Why does soap not lather well in hard water?

Lesson 4 · Fuels and flames (about 9 min)

Fuel	Calorific value (kJ/g, approx.)	Note
Hydrogen	≈ 150 (highest)	Clean; hard to store
LPG	≈ 55	Mainly butane (+ propane); smell added (ethyl mercaptan) to detect leaks
CNG	≈ 50	Mainly methane; cleaner than petrol and diesel
Petrol, kerosene, diesel	≈ 45	Petroleum fractions
Coal	≈ 25–33	Peat → lignite → bituminous → anthracite (most carbon)
Wood · cow-dung cake	≈ 17–22 · ≈ 6–8	Smoky

Biogas, marsh gas and CNG are mainly methane. Coal gives coke, coal tar and coal gas. Petroleum refining gives LPG, petrol, kerosene, diesel, lubricating oil, paraffin wax and bitumen.

Candle flame: outermost zone — complete combustion, **hottest**, non-luminous (goldsmiths use it); middle — partial combustion, luminous; innermost — unburnt wax vapour, least hot.

Fire safety: CO_2 is best for electrical and oil fires; never water on them.

Quick check 4 — answer out loud before moving on (answers at the end)

a) Main component of CNG? b) Hottest zone of a candle flame? c) Which coal has the most carbon?

Lesson 5 · Polymers, glass, cement and medicines (about 8 min)

Material	Fact
Polythene · PVC	Plastics; PVC from vinyl chloride (pipes)
Teflon (PTFE)	Non-stick cookware
Nylon	First fully synthetic fibre
Rayon	'Artificial silk', made from wood pulp (cellulose)
Bakelite, melamine	Thermosetting plastics — don't soften on heating (switches, crockery)
Natural rubber	Polymer of isoprene; vulcanisation with sulphur (Goodyear) makes it hard
Glass	Sand (SiO_2) + Na_2CO_3 + CaCO_3 · cobalt oxide gives blue glass
Cement	Portland cement; gypsum added to slow setting

Medicine type	Does what	Example
Analgesic	Relieves pain	Aspirin, paracetamol
Antipyretic	Lowers fever	Paracetamol
Antibiotic	Kills or stops bacteria	Penicillin (Alexander Fleming, 1928)
Antacid	Neutralises excess stomach acid	Milk of magnesia $\text{Mg}(\text{OH})_2$
Antiseptic	Used on living tissue	Dettol, tincture of iodine, boric acid

Quick check 5 — answer out loud before moving on (answers at the end)

a) Polymer used to coat non-stick pans? b) Why is gypsum added to cement?

Lesson 6 · Common names and formulas (high-yield table) (about 9 min)

Common name	Formula	Common name	Formula
Quicklime	CaO	Slaked lime	$\text{Ca}(\text{OH})_2$
Limestone / marble / chalk	CaCO_3	Caustic soda	NaOH
Laughing gas	N_2O	Dry ice	Solid CO_2
Heavy water (reactor moderator)	D_2O	Oil of vitriol	H_2SO_4
Muriatic acid	HCl	Green vitriol	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$
White vitriol	$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$	Epsom salt	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
Potash alum (purifies water)	$\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$	Sal ammoniac	NH_4Cl
Saltpetre (gunpowder)	KNO_3	Chile saltpetre	NaNO_3
Urea ($\approx 46\%$ N)	NH_2CONH_2	Philosopher's wool	ZnO
Cane sugar	$\text{C}_{12}\text{H}_{22}\text{O}_{11}$	Marsh gas	CH_4

Quick check 6 — answer out loud before moving on (answers at the end)

a) Common name of N_2O ? b) Formula of slaked lime? c) What is heavy water used for?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Carbon	Tetravalent, catenation · alkane C_nH_{2n+2} , alkene C_nH_{2n} , alkyne C_nH_{2n-2} · acetylene welding · ethene ripens fruit
Functional groups	-OH alcohol · -CHO aldehyde (formalin) · ketone (acetone) · -COOH acid (vinegar 5-8% acetic) · ester sweet smell
Reactions	Hydrogenation needs nickel → vanaspati · saponification → soap · methanol → blindness
Soap	Micelles · hard water Ca^{2+}/Mg^{2+} → scum · detergents work in hard water · temporary hardness boiled, permanent → washing soda
Fuels	H_2 highest calorific value · LPG butane · CNG/biogas/marsh gas methane · anthracite most carbon · outer flame hottest
Polymers	Teflon non-stick · nylon first synthetic · rayon from wood pulp · Bakelite thermosetting · vulcanisation with sulphur
Medicines	Paracetamol analgesic + antipyretic · penicillin Fleming · antacid $Mg(OH)_2$ · antiseptic Dettol, iodine
Names	N_2O laughing gas · D_2O heavy water · H_2SO_4 oil of vitriol · KNO_3 saltpetre · alum purifies water

Science practice set — 20 questions · target 6 minutes

#	Question
1	The property of carbon to form long chains is called: (a) allotropy (b) isomerism (c) catenation (d) hybridisation
2	The general formula of alkanes is: (a) C_nH_{2n} (b) C_nH_{2n+2} (c) C_nH_{2n-2} (d) C_nH_n
3	Which gas is used with oxygen for welding? (a) Methane (b) Acetylene (ethyne) (c) Ethene (d) Butane
4	Vinegar is a dilute solution of: (a) citric acid (b) formic acid (c) lactic acid (d) acetic acid
5	Formalin (used to preserve specimens) is a solution of: (a) ethanol (b) formaldehyde (c) acetone (d) acetic acid
6	Drinking which alcohol can cause blindness? (a) Methanol (b) Ethanol (c) Glycerol (d) Propanol
7	Hydrogenation of vegetable oil uses the catalyst: (a) iron (b) platinum (c) nickel (d) copper
8	Unsaturated hydrocarbons burn with a: (a) clean blue flame (b) green flame (c) sooty yellow flame (d) no flame
9	Soaps do not work well in hard water because of: (a) sodium ions (b) chloride ions (c) dissolved oxygen (d) calcium and magnesium ions
10	Making soap from fat and NaOH is called: (a) esterification (b) saponification (c) hydrogenation (d) fermentation
11	The fuel with the highest calorific value is: (a) LPG (b) hydrogen (c) petrol (d) coal
12	CNG is mainly: (a) butane (b) propane (c) ethane (d) methane
13	The main component of LPG is: (a) methane (b) ethane (c) hydrogen (d) butane
14	Which coal has the highest carbon content? (a) Anthracite (b) Peat (c) Lignite (d) Bituminous
15	The hottest zone of a candle flame is the: (a) innermost zone (b) middle zone (c) wick (d) outermost zone
16	Teflon is used for: (a) electric switches (b) tyres (c) artificial silk (d) non-stick cookware
17	Rayon is made from: (a) wood pulp (cellulose) (b) petroleum (c) coal tar (d) silk worms
18	Gypsum is added to cement to: (a) speed up setting (b) add colour (c) increase strength only (d) slow down setting
19	Laughing gas is: (a) NO_2 (b) N_2O (c) NO (d) N_2O_5
20	Heavy water is used in nuclear reactors as a: (a) fuel (b) coolant only (c) shield (d) moderator

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	c	Lesson 1
2	b	Lesson 1
3	b	Lesson 1
4	d	Lesson 2
5	b	Lesson 2
6	a	Lesson 2
7	c	Lesson 3
8	c	Lesson 3
9	d	Lesson 3
10	b	Lesson 3
11	b	Lesson 4
12	d	Lesson 4
13	d	Lesson 4
14	a	Lesson 4
15	d	Lesson 4
16	d	Lesson 5
17	a	Lesson 5
18	d	Lesson 5
19	b	Lesson 6
20	d	Lesson 6

Quick checks

QC	Answers
1	a) C_nH_{2n} b) Acetylene (ethyne)
2	a) Acetic (ethanoic) acid, about 5–8% b) $-OH$ (hydroxyl)
3	a) Nickel b) Calcium and magnesium ions in hard water react with soap to form scum
4	a) Methane b) The outermost (non-luminous) zone c) Anthracite
5	a) Teflon (PTFE) b) To slow down setting
6	a) Laughing gas b) $Ca(OH)_2$ c) As a moderator in nuclear reactors

Day 11 recall (Block 0)

#	Answer	#	Answer
1	Mercury	6	Bauxite
2	Gold	7	Germanium
3	Silver	8	Argon
4	Copper + zinc	9	Halogens
5	Weld railway tracks	10	2 : 3

Send me tonight

Day 11 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.